Users start to give feedback that they would love a way to find contacts using *typeahead* (the application gives suggestions while the user types part of a contact's name) and automatically receive suggestions for teams and current conversations based on their recent email conversations. You build a `TeamFinder` object that calls out to several web services like the Google Contacts API and Microsoft Outlook.com API. You build these web service clients and incorporate the finding of contacts, as shown in figure 1.8.

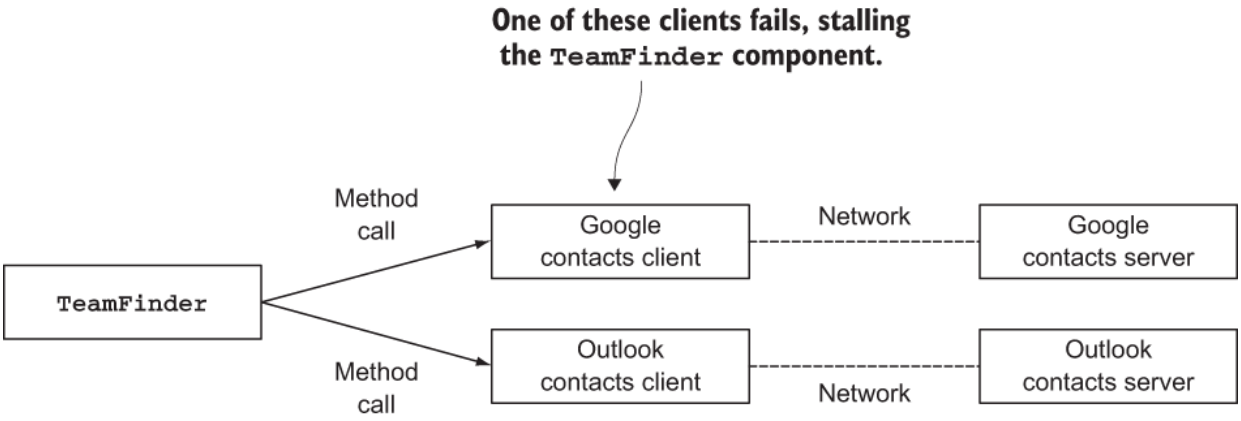


Figure 1.8 `TeamFinder` component

Then you find out that one of the services often fails in the worst possible way: you get long timeouts, or traffic slows to only a few bytes per minute. And because the web services are accessed one after the other, the `TeamFinder`—waiting for the response—fails the lookup after a long time even though the service that worked fine could have made many valid suggestions to the user.

Even worse, although you have collected your database methods in DAOs and the contact lookup in a `TeamFinder` object, the controllers call these methods like any others. This means a user lookup sometimes happens between two database methods, keeping